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Cake Holding Robotic Icing System To Make A Snack

Table of Contents

Table of Contents.....	2
Design.....	2
The Base mechanism.....	4
The Syringe mechanism.....	4
The X-axis Mechanism.....	5
Design highlights and difficulties.....	6
Derivation of Kinematics.....	7
Control Considerations.....	8
Workspace Analysis.....	9
Task Simulation.....	11
Performance comparison to commercial products.....	11

Design

Our goal was to design a cake-icing robot capable of icing both the top and vertical sides of a cylindrical cake. Achieving full coverage requires coordinated translational and rotational motions. For top icing, we needed either an X–Y gantry or an X-axis translation combined with rotation of the cake; for side icing, we additionally required a Z-axis motion and a way to trace patterns around the perimeter.

Because the cake is cylindrical, we chose to rotate the cake rather than build a full X–Y system, simplifying the design and reducing the actuator count. The base platform therefore provides two DOFs: vertical translation (Z-axis) and rotation. With rotation already available, the top surface can be iced using only a single X-axis motion at the nozzle, allowing various patterns through coordinated X + rotation control.

To ice the sides, we added a rotational nozzle so the end effector can orient at the required angle, and we incorporated a controlled syringe-extrusion mechanism to synchronize icing flow with the X, Z, rotation, and nozzle-orientation motions.

Meeting these requirements leads to seven motors: three for Z-axis lifting, one for cake rotation, one for X translation, one for end-effector rotation, and one for extrusion. The three Z motors together act as one DOF, giving the robot a total of five DOFs.

Figure 1 shows the simulated concept and the final assembled robot; the next sections describe each subsystem.

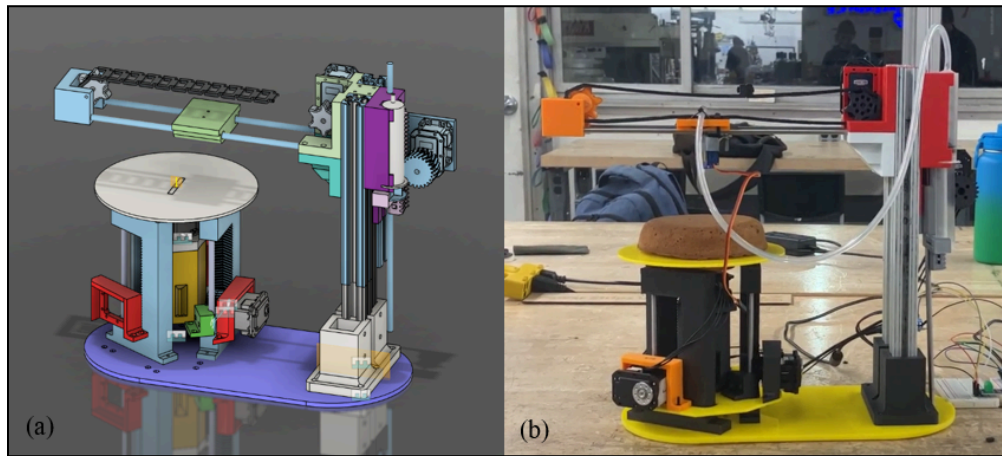


Figure 1. (a) Simulation of the robot concept. (b) Completed robot after 3D-printed components were fabricated and assembled.

The Base mechanism

The base mechanism houses four motors: three for Z-axis lifting of the cake and its carriage, and one directly beneath the cake to provide continuous rotation. The rotation system was simple—a motor mounted under a connector and a laser-cut circular plate. The connector was fastened to the motor shaft and bonded to the plate, enabling smooth rotation of the cake.

For the Z-axis motion, we evaluated lead screws, ball screws, and rack-and-pinion systems. Since the parts were to be 3D-printed and needed to remain lightweight and tolerant of small dimensional errors, we selected a rack-and-pinion design. Custom racks with mounting extrusions were created to attach to the base plate (Fig. 2a), and three steel guide rods were added to improve rigidity and reduce flex during lifting.

We then designed a hexagonal carriage that holds the three Z motors, mates each to its pinion gear, and slides along the guide rods using linear bearings. The carriage geometry ensures symmetric loading and even lifting. Central screw holes allow attachment of a 3D-printed extrusion linking the carriage to the fourth motor that rotates the top plate. The laser-cut bottom plate includes matching screw holes for securing the racks and the aluminum-extrusion holders, completing the structural support of the base.

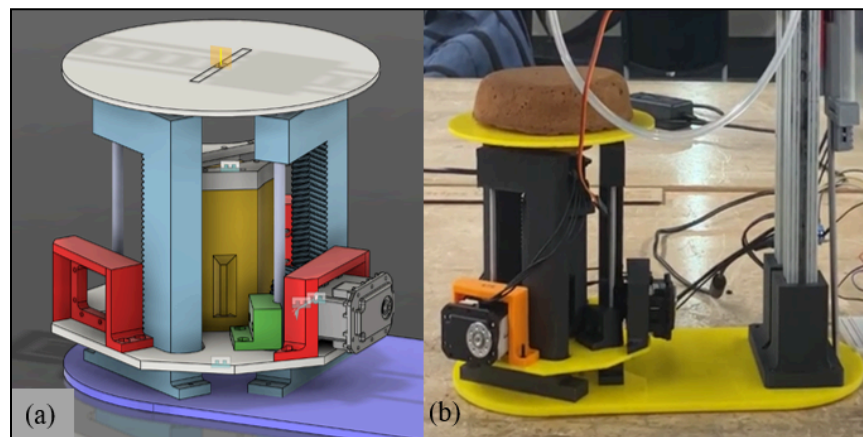


Figure 2. (a) Simulation of the base mechanism to move the cake up and rotate it (b) Completed base mechanism after 3D-printed components were fabricated and assembled.

The Syringe mechanism

We selected a standard syringe, modeled it in CAD, and designed the surrounding components accordingly. A syringe holder was created to mount onto the one-foot aluminum extrusion using screws and T-nuts, providing stability, adjustability, and slots for securing the motor. Because the motor outputs rotational motion, we needed a compact way to convert it into linear displacement

to drive the plunger. Due to space constraints, we chose a rack-and-pinion system, designing the rack with an extended tab that directly connects to the syringe plunger for smooth linear travel.

To reduce slipping and backlash under load, we used a double-angled tooth profile, improving engagement and ensuring consistent icing extrusion. The final simulation and assembled syringe mechanism appear in Figure 3.

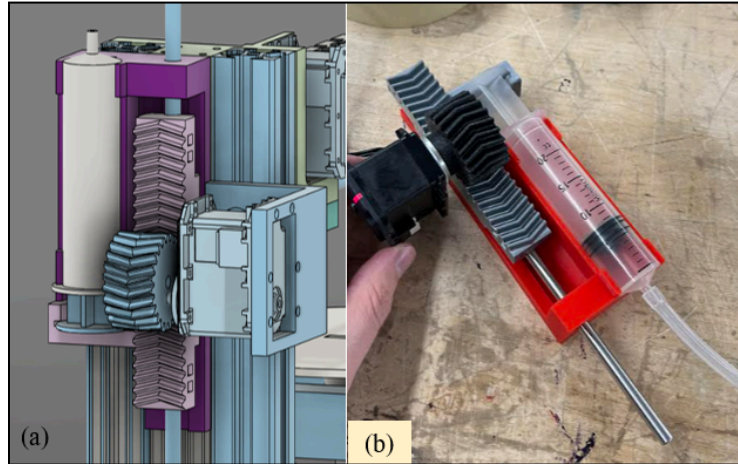


Figure 3. (a) Simulation of the rack-and-pinion assembly and the slot designed to drive the syringe plunger and extrude the icing. (b) Completed syringe mechanism after fabrication and assembly of all 3D-printed components.

The X-axis Mechanism

For the final part of the design, we created the X-axis translation above the cake, again converting motor rotation into linear motion. We implemented a 3D-printed belt-and-pinion system with two pinions: an idler freely rotating on a circular rod and a driven pinion attached to the motor shaft. A mounting assembly was designed to attach to the aluminum extrusion, hold the motor, and provide slots for the X-axis guide rails, with a second component supporting the opposite rail ends and the idler.

To maintain proper vertical alignment with the Z-axis motion, a tubular element from the Z carriage was connected to the X carriage, ensuring a constant nozzle height—while the belt drove the horizontal slider along the rails. The final motor was mounted on the X carriage with a cylindrical attachment through which the tube passed, allowing rotation of the end effector for adjustable icing orientation. The finished X-axis mechanism is shown in Figure 4.

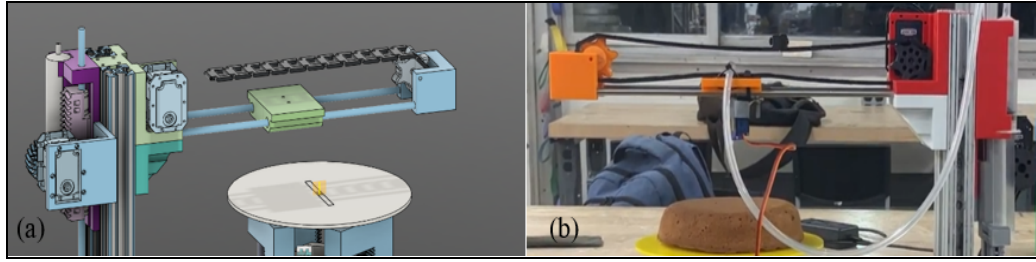
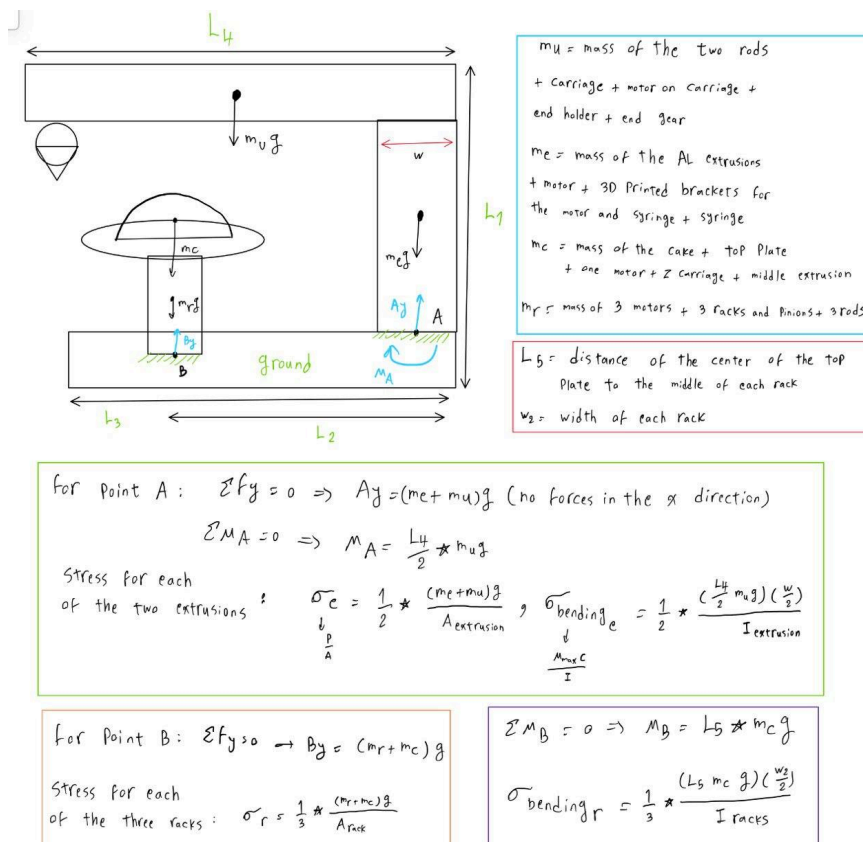


Figure 4. (a) Simulation of the X axis assembly and the belt designed to move the end effector to the desired locations. (b) Completed X axis mechanism after fabrication and assembly of all 3D-printed components.

Design highlights and difficulties

One of the major challenges in this project was finding reliable ways to convert rotational motion into translational motion three different times within tight spatial and time constraints. Designing the Z-axis carriage was particularly demanding, as it needed to hold all three vertical motors, provide clearance for the racks, accommodate the central 3D-printed extrusion, and still travel smoothly without collisions. The X-axis system also posed difficulties: because the entire assembly moves along only two rods and a belt, we needed creative structural solutions to ensure stable attachment and controlled motion of all components. Despite these complexities, we successfully developed functional mechanisms through iterative brainstorming and effective teamwork.

Static force analysis



Derivation of Kinematics

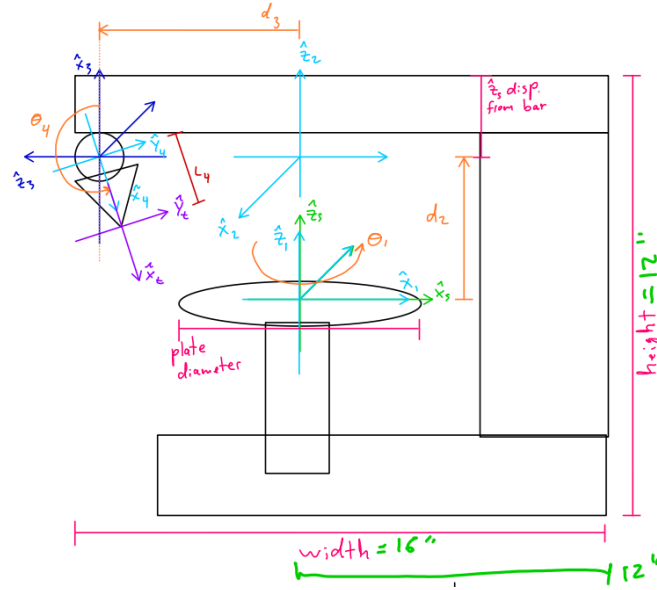


Figure 5: Diagram of robot with DH frames attached

$i - 1$	i	α_{i-1}	a_{i-1}	θ_i	d_i
S	1	0	0	θ_1	0
1	2	0	0	-90°	d_2
2	3	90°	0	90°	d_3
3	4	90°	0	θ_4	0
4	T	0	L_4	0	0

Table 1: DH Parameters. Note that L_4 can change easily depending on how the tube was placed during robot setup. Simulations were done with $L_4 = 3$ cm.

Given these parameters, the following describes the forward kinematics:

$${}^0_1T = \begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1_2T = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^3_4T = \begin{pmatrix} \cos(\theta_4) & -\sin(\theta_4) & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \sin(\theta_4) & \cos(\theta_4) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^4_TT = \begin{pmatrix} 1 & 0 & 0 & L_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^S_T T = {}^S_1 T {}^1_2 T {}^2_3 T {}^3_4 T {}^4_T T$$

$${}^S_T T = \begin{pmatrix} -\cos(\theta_1) \sin(\theta_4) & -\cos(\theta_1) \cos(\theta_4) & \sin(\theta_1) & -d_3 \cos(\theta_1) - L_4 \cos(\theta_1) \sin(\theta_4) \\ -\sin(\theta_1) \sin(\theta_4) & -\sin(\theta_1) \cos(\theta_4) & -\cos(\theta_1) & -d_3 \sin(\theta_1) - L_4 \sin(\theta_1) \sin(\theta_4) \\ \cos(\theta_4) & -\sin(\theta_4) & 0 & d_2 + L_4 \cos(\theta_4) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Using geometric derivation techniques, as well as the cartesian coordinate equations given in the tool to station transfer matrix, the following describe the inverse kinematics (note that inverse tangents are implemented using atan2, thus supporting denominators equal to zero).

$$\theta_1 = \tan^{-1} \left(\frac{y}{x} \right)$$

$$d_2 = z - L_4 \cos(\theta_4)$$

$$d_3 = \begin{cases} -\frac{x + L_4 \cos(\theta_1) \sin(\theta_4)}{\cos(\theta_1)} & \cos(\theta_1) \neq 0 \\ -\frac{y + L_4 \sin(\theta_1) \sin(\theta_4)}{\sin(\theta_1)} & \sin(\theta_1) \neq 0 \end{cases}$$

$$\theta_4 = \theta_{ex}$$

Because cake is often cylindrical, it makes sense to derive the inverse kinematics given a cylindrical coordinate system as well. This yields the following.

$$\theta_1 = \theta_1$$

$$d_2 = z - L_4 \cos(\theta_4)$$

$$d_3 = r - L_4 \sin(\theta_4)$$

$$\theta_4 = \theta_{ex}$$

Note that solutions uniquely describe the position and orientation of the end effector, so solution selection is not necessary, as there is only ever one solution.

Control Considerations

Since drawing straight lines on a cake would be difficult using a rotary platform without high resolution trajectory points, velocity-based motor control is a useful tool. So, the derivatives of the inverse kinematic equations are taken to get the rates of change of the DH variables, given both desired trajectory states and rates.

$$\dot{\theta}_1 = \begin{cases} \frac{x\dot{y}-y\dot{x}}{x^2+y^2} & \text{otherwise} \\ 0 & x = y = 0 \end{cases}$$

$$\dot{d}_2 = \dot{z} + \dot{\theta}_4 L_4 \sin(\theta_4)$$

$$\dot{d}_3 = \begin{cases} \frac{-\dot{x} + \dot{\theta}_1 (L_4 s_1 c_1 - (x + L_4 c_1 s_4) t_1) - L_4 \dot{\theta}_4 c_1 c_4}{c_1} & c_1 \neq 0 \\ \frac{-\dot{y} + \dot{\theta}_1 \left((y + L_4 s_1 s_4) \frac{1}{t_1} - L_4 c_1 s_4 \right) - L_4 \dot{\theta}_4 s_1 c_4}{s_1} & s_1 \neq 0 \end{cases}$$

$$\dot{\theta}_4 = \dot{\theta}_{ex}$$

Once again, this is also done in polar coordinates.

$$\dot{\theta}_1 = \dot{\theta}_1$$

$$\dot{d}_2 = \dot{z} + L_4 \dot{\theta}_4 \sin(\theta_4)$$

$$\dot{d}_3 = \dot{r} - L_4 \dot{\theta}_4 \cos(\theta_4)$$

$$\dot{\theta}_4 = \dot{\theta}_{ex}$$

Workspace Analysis

Setting the turntable as our fixed reference frame simplifies our definition for the workspace. Looking at the figure below, the maximum range of the extruder in the radial axis, from the center axis is $R_{rb} = 5$ in, $R_{ru} = 4$ in, $R_{lb} = 3$ in and $R_{lu} = 4$ in.

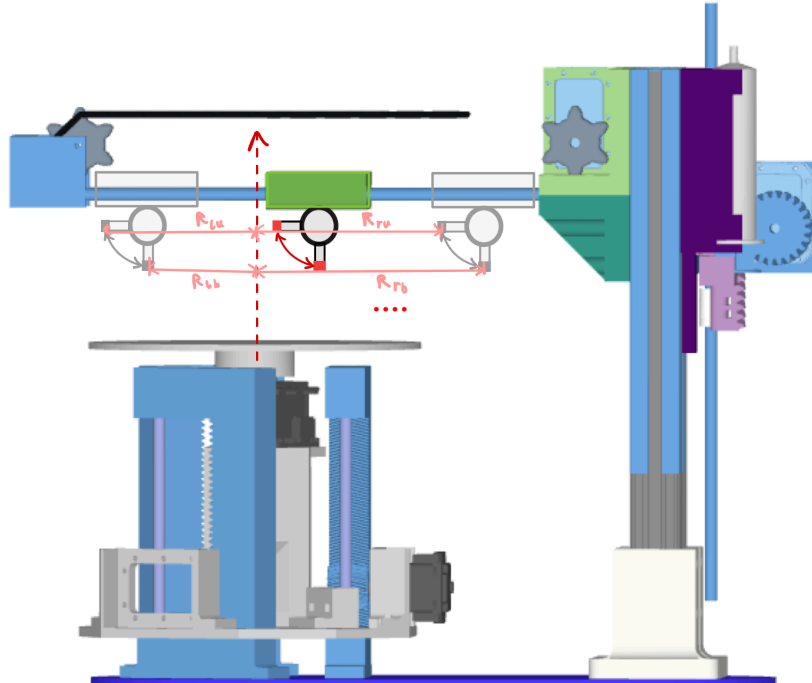


Figure 6: Reachable workspace of extruder in context of the robot

Since the extruder is capable of sweeping around the turntable, the sweep created by the leftmost position of the extruder is a subset of the sweep created by the extruder at the rightmost position. The figures below, provides the reachable workspace for a fixed turntable angular position.

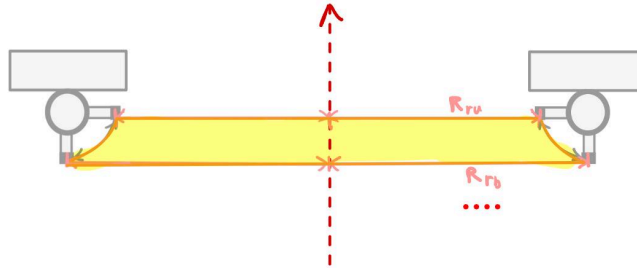


Figure 7: Reachable workspace using 2 DOF (radial and extruder orientation)

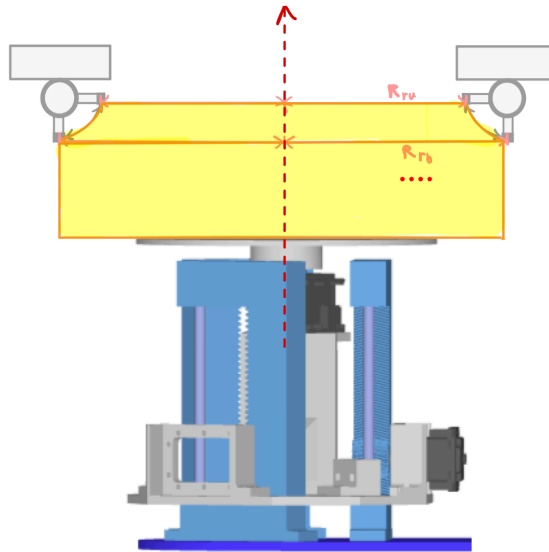


Figure 8: Reachable workspace using 3 DOF (radial, z-axis and extruder orientation)

Figure 8 provides the workspace reached by the extruder when the turntable platform is allowed to rise and fall. By adding the angular rotation of the turntable, we end up with a workspace that is cylindrical. Since the end extruder only has 90 degrees as its range of motion, there is no Dexterous workspace.

Task Simulation

In order to test the math for the cartesian and polar coordinates, separate tests were done. The polar simulation draws a spiral on top of the cake, followed by waves on the side of the cake. This also tests all 4 degrees of freedom. The cartesian simulation writes the word “HI” on the top of the cake. The results are found below.

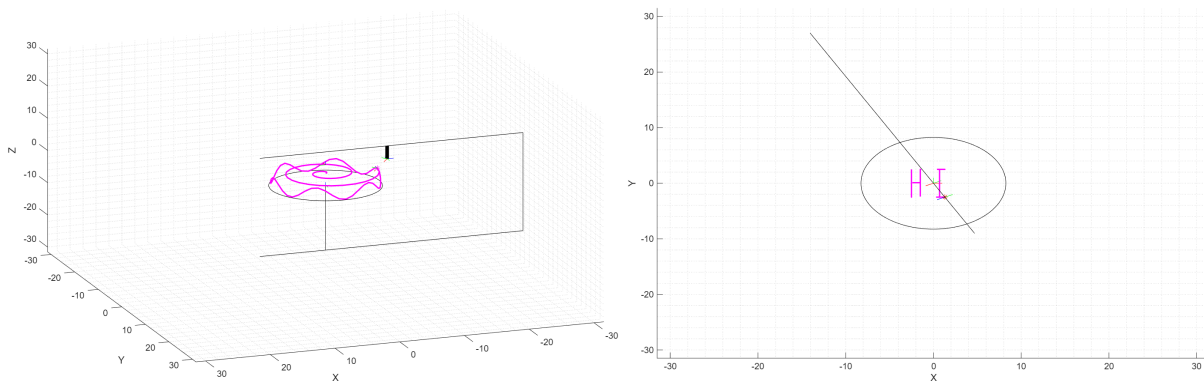


Figure 9: Polar coordinate (left) and Cartesian coordinate (right) simulations. Note that the L4 value for the simulations was set to 3cm.

Performance comparison to commercial products

Our bot is very similar to a 3D printer with a rotating nozzle, as it has an extruder and a build plate with motors providing it with 4 Degrees of Freedom. These printers are extremely robust and capable of achieving high acceleration while preventing any disturbances. There is also the commercially available Cakewalk 3D, which is an extruder mod that can be attached to an existing 3D printer. This is an elegant solution that reuses the functionality of a printer and the control systems used to control the motors. But, this model can only apply icing on a flat surface while varying the height, while our robot is able to account for the contours of the cake.

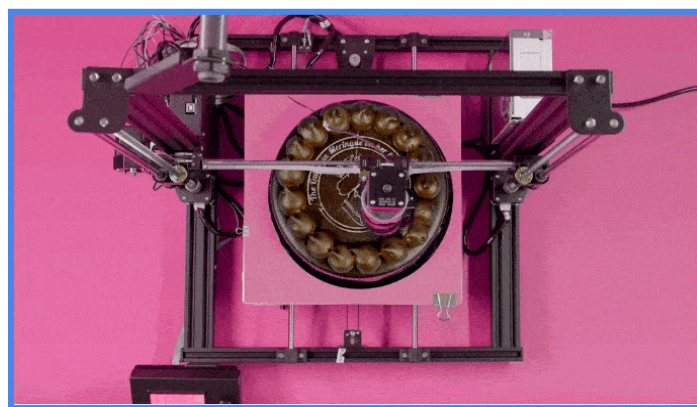


Figure 10: Application of icing using Cakewalk 3D